Topic 16: Proof Techniques in Economic Theory

(Making conscious what you have been doing all along)

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1. Why this topic exists

You have now read, and in places worked through, a substantial number of mathematical arguments. The IVT was proved by chasing the supremum of a sublevel set (Topic 4). Brouwer in one dimension was reduced to the IVT; in higher dimensions the argument proceeded by assuming no fixed point and deriving a geometric impossibility (Topic 10). Kakutani was proved by constructing a sequence of ordinary functions that approximate the correspondence, applying Brouwer to each, and passing to a limit (Topic 10). The supporting hyperplane was found by approaching a boundary point from outside, projecting at each step, and extracting a convergent subsequence (Topic 13). The Lebesgue integral was built by summing horizontal bricks of width ε and letting $\varepsilon \to 0$ (Topic 15).

Each of these arguments worked. But at no point were you told *why* a particular strategy was chosen, or how you would have known to try it if you were starting from scratch. The goal of this topic is to fill that gap.

We will give names to the techniques you have already seen, identify the situations in which each technique is the right one, and look across the whole course to find the recurring *patterns* that economic theorems follow. The goal is not to turn you into a logician. It is to make you a more deliberate reader and writer of proofs: to give you a small number of named strategies to try when you are stuck, and a sense for which economic structures call for which strategies.

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2. What proofs actually do in economics

There is a standard line about proofs in mathematics: they establish truth rigorously. This is obviously important, and one of the reasons that separate "vibe" arguments from (at least) internally consistent ones. Moreover, given the degrees of freedom given to us in modeling economics, there is another reason for a proof, especially in theory.

A proof tells us which assumptions are doing what work. This is not the same as establishing a conclusion. A statement establishes a non-trivial causal relationship: *if* these conditions hold, *then* this conclusion follows. The proof is the chain that runs from the conditions to the conclusion, delivering (ideally) an elegant line of reasoning.

Consider the Weierstrass theorem (Topic 4): a continuous function on a compact set attains its maximum. The proof uses compactness to extract a convergent subsequence, then uses continuity to pass the limit through f. If you remove compactness, the argument fails at the first step (there is no convergent subsequence). If you remove continuity, it fails at the second step (you cannot pass the limit through f). The proof therefore tells you something economically meaningful: compactness is what prevents the optimum from escaping to infinity or to the boundary of the domain, and continuity is what ensures the objective value is preserved in the limit. Neither assumption is there without a reason.

The same logic runs through every theorem in the course. Kakutani needs convex values because the approximating functions are constructed by averaging, and averaging only stays inside the correspondence's image if those images are convex (Topic 10). The contraction mapping theorem needs $\beta < 1$ because that is precisely the condition that makes the geometric series $\sum \beta^n$ converge and drives the Cauchy estimate to zero (Topic 12). The separating hyperplane theorem needs the sets to be convex because the closest-point projection only behaves as required for convex sets (Topic 13).

This is why proofs matter. They are the only reliable answer to the question "what happens if I weaken this assumption?" If you have a proof, you can look at which steps use the assumption and ask whether they can be rescued. The statement alone (even if correct) does not tell you that.

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3. The basic toolkit

Every proof in this course—and in most of microeconomic theory—uses one or more of four elementary techniques. We describe each one, identify where it appeared in the course, and explain why it was the right choice in each case.

3.1 Direct proof

A **direct proof** starts from the hypotheses and constructs the conclusion step by step, using definitions, previously established results, and algebraic manipulation. There is no auxiliary assumption, no case that is supposed to be impossible, just a chain of deductions.

Most of the algebraic work in the course is direct proof. When we checked that the Bellman operator is a contraction (Topic 12), the argument was: write out $\|TW - TW'\|_\infty$, substitute the definition of T, use the fact that $|W(\theta) - W'(\theta)| \leq \|W - W'\|_\infty$ everywhere, and conclude that $\|TW - TW'\|_\infty \leq \delta \|W - W'\|_\infty$. Each line follows from the previous one by direct substitution or inequality. No detour through an impossible case, no side assumption that is later discarded. You start at the hypotheses and walk straight to the conclusion.

When is direct proof the right strategy? Almost always, when you can see a path from the hypotheses to the conclusion. The challenge is that "seeing the path" often requires knowing what objects to introduce along the way. In the Bellman operator argument, the key object was the sup norm $\|\cdot\|_\infty$, and the key move was bounding the pointwise difference uniformly. Developing an instinct for these intermediate objects is most of what it means to be good at direct proofs.

The IVT proof in Topic 4 is a more subtle example of a direct proof. You want to show a particular value is attained. The intermediate object introduced is the set $X = \{x \in [a, b] : f(x) \leq c\}$, whose supremum $\hat{x}$ turns out to be the point you want. The argument then rules out $f(\hat{x}) < c$ and $f(\hat{x}) > c$ by continuity, and concludes $f(\hat{x}) = c$ by elimination. Every step is direct deduction; there is no assumption that is later contradicted.

Direct proofs are constructive but not necessarily pedagogical. They tell us step by step how to move from assumptions to results—that is the constructive part. They are also sometimes lengthy and cumbersome to read. Algebra delivers only so much intuition—that is the more difficult part.

3.2 Proof by contradiction

A **proof by contradiction** assumes that the conclusion is *false*, and then derives a consequence that contradicts the hypotheses or a known fact. The logical structure is: if the conclusion were false, then something impossible would follow; therefore the conclusion must be true. The ideas (like the proof by contrapositive below) is based on the logical identity that if the statement P implies Q is true, that means that if Q is not true, then also P cannot be true. What the proof by contradiction does, is to say, suppose Q is falls (it is $\not Q$), can we find something in P to still generate our element in $\not Q$. If so, the statement cannot be true. If any such attempts leads us outside the set P, we cannot get there.

The higher-dimensional proof of Brouwer's theorem is the clearest example in the course (Topic 10). We want to show: every continuous self-map of B^n has a fixed point. We assume the opposite—that f has *no* fixed point—and ask what that implies. It implies we can draw a ray from $f(x)$ through x for every x, which defines a retraction $r : B^n \to S^{n-1}$ (a continuous map that keeps the boundary sphere fixed). But Borsuk's theorem says no such retraction can exist. Contradiction. Therefore our assumption was wrong, and a fixed point must exist.

Notice the structure. The conclusion ("a fixed point exists") is an *existence claim*, and existence claims are often hard to prove directly: you would need to construct the fixed point explicitly. But the negation ("no fixed point exists") is a universal claim ("for all x, $f(x) \neq x$"), and universal claims are often easier to work with because they give you a property that holds everywhere. That universal property—the existence of the ray at every point—is exactly what builds the retraction, and it is the retraction that leads to the contradiction.

A second example is the uniqueness argument in the contraction mapping theorem (Topic 12). We want to show there is at most one fixed point. We assume there are two, x^* and y^*. Then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq \beta\, d(x^*, y^*)$. Since $\beta < 1$ and $d(x^*, y^*) \geq 0$, this forces $d(x^*, y^*) = 0$, hence $x^* = y^*$. Contradiction.

The separating hyperplane proof also uses contradiction in an embedded form (Topic 13, Step 0). To show the closest point z separates y from C, we suppose some $x \in C$ is on the wrong side ($p \cdot x > p \cdot z$) and show that the midpoint between z and x would then be closer to y than z is—contradicting the minimality of z.

Proof by contradiction is often short and elegant. You show that if the statement was not true, then any construction involves that you fall out of the set of assumptions. However, the proof by contradiction is by construction not constructive (pun intended!). You only learn that something has to be true, not why. There is no chain of arguments delivering the results. Instead there is a chain of arguments that excludes other options.

3.3 Proof by contrapositive

The **contrapositive** of a statement "if P then Q" is "if not Q then not P." The two are logically equivalent, so proving the contrapositive proves the original statement. This is in somesense the direct version of a proof by contradiction. But instead of just showing we always end up in a contradiction, but we show why the contrapositive must be true in a constructive manner, that is, there is more to learn as to why at least the contrapositive is as it is. Contrapositive is most useful when the forward direction $P \Rightarrow Q$ is hard to see directly, but the reverse direction "no Q implies no P" has an obvious witness.

Contrapositive proofs are less prominent in the course than contradiction, but the logic underlies several arguments. Consider: "if the Bellman operator has $\delta \geq 1$, it need not be a contraction." This is the contrapositive framing of the necessity of $\delta < 1$: the assumption $\delta < 1$ is not just sufficient for a contraction, it is (essentially) necessary. The counterexamples that appear at the end of the convergence theorem section in Topic 15—-sequences that fail to converge when one hypothesis is dropped—-are all instances of contrapositive reasoning: they exhibit "not Q" (convergence fails) when "not P" (one hypothesis is violated), thereby confirming that P is genuinely needed.

When reading a theorem statement, it is always worth asking: "what is the contrapositive?" If the theorem says "compactness implies existence of a maximum," the contrapositive says "if the maximum does not exist, the set was not compact (or the function was not continuous)." This reframing often gives a clean intuition for *why* the hypothesis is needed.

3.4 Mathematical induction

Mathematical induction proves a statement $P(n)$ for all natural numbers n by: (i) verifying one version (typically for the starting value $n = 0$ or $n = 1$), and (ii) showing that if $P(n)$ holds then $P(n + 1)$ holds (the inductive step). The logic is that if the first domino falls and each domino knocks over the next, all dominoes fall.

Induction appears most naturally in the course in the analysis of geometric series

(Topic 11). To show that $\sum_{k=0}^{n} \delta^k = (1-\delta^{n+1})/(1-\delta)$, you check it for $n = 0$ (both sides equal 1), then assume it holds for n and add the $(n+1)$-th term to both sides. More substantially, the contraction mapping proof (Topic 12) is an inductive argument in disguise: the estimate $d(x_n, x_{n+1}) \leq \beta^n\, d(x_0, x_1)$ is proved by applying the contraction property n times – one application per step of the iteration – which is exactly the inductive structure.

Induction is also the right technique for arguing about *finitely iterated* procedures: backward induction in dynamic games, iterated elimination of dominated strategies, finite-period dynamic programs. Any time a result "propagates backwards" from a known endpoint, the formal justification is induction.

3.5 A brief note on "proof by cases"

Sometimes an argument splits into cases because the conclusion holds for different reasons depending on which case you are in. The IVT proof handles the case $f(\hat{x}) < c$ and $f(\hat{x}) > c$ separately. The supporting hyperplane proof handles $\hat{x} = a$, $\hat{x} = b$, and $\hat{x} \in (a, b)$ as sub-cases. This is not a technique in the same sense as the above—it is more of a bookkeeping device. The key discipline is: cover every case, and do not let cases overlap in a way that lets you avoid a hard sub-argument.

That said, cases are often prevalent for general statements because sometimes things hold in different regions, but for different reasons. To see why, it makes sense to break up the statement and prove it case-by-case.

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4. Construction and approximation

Two techniques that have a distinctive flavor in analysis and in economic theory deserve separate treatment.

4.1 Construction proofs

A **construction proof** establishes existence by explicitly building the object whose existence is claimed. It is the most direct of all existence arguments: rather than assuming non-existence and deriving a contradiction, you point to the object and verify it has the required properties.

Tarski's fixed-point theorem (Topic 10) is proved constructively. The claimed fixed point is $x^* = \sup\{x \in X : x \preceq f(x)\}$ – the supremum of the set of points "below their image." You then verify: $f(x^*)$ is an upper bound for that set (so $x^* \preceq f(x^*)$), and $f(x^*)$ is itself in the set (so $f(x^*) \preceq x^*$), giving $x^* = f(x^*)$. The fixed point is not found by ruling out alternatives; it is built from the lattice structure and handed to you.

The Lebesgue integral (Topic 15) is constructed similarly. You want to define $\int f\, d\mu$ for a nonnegative measurable function. Step one: define the integral for step functions (finite sums of indicator functions over measurable sets), where it is just a weighted sum of measure times height. Step two: define the integral for a general nonneg function as the supremum of the integral over all step functions that lie below it. The construction works because the lattice structure of the real line ensures the supremum is well-defined and the monotone convergence theorem guarantees it is stable under limits. You never "prove the integral exists by contradiction"—you build it, brick by brick, from simpler objects.

Why construction? Because it delivers a clear intuition what is happening.[1]

4.2 Approximation-then-limit

An **approximation-then-limit** proof proceeds in three steps:

1. Replace the object of interest (which is complicated or infinite-dimensional) with a sequence of simpler approximations.
2. Prove the desired result for each approximation.
3. Show that the limit of the approximations inherits the result.

The transition from step 2 to step 3 is always the hard step, and it always requires a *compactness* or *closedness* argument that ensures the limit is well-behaved.

The Kakutani proof (Topic 10) is the clearest example. The correspondence F is complicated – it is set-valued, and we cannot apply Brouwer directly. The approximating objects are piecewise linear functions f_n that "fill in" the correspondence on a grid of mesh $1/n$. Each f_n is an ordinary continuous function, and Brouwer gives a fixed point $x_n = f_n(x_n)$. Compactness of X ensures the sequence (x_n) has a convergent subsequence with limit x^*. The closed graph property of F (which follows from upper hemicontinuity and compact values – recall Topic 5) then ensures that the limit point satisfies $x^* \in F(x^*)$. The crucial phrase: *each* x_n is only an *approximate* fixed point of F, and it is the closed graph that converts "approximate" into "exact." Without that final ingredient, the approximation argument only gives you a sequence – not a conclusion.

The supporting hyperplane proof (Topic 13) follows the same template. The object of interest is the normal vector to a hyperplane tangent to C at a boundary point x^*. The approximations are the normal vectors $p_k = (y_k - z_k)/\|y_k - z_k\|$ obtained by projecting points $y_k \to x^*$ from outside C. Each p_k is a unit vector, so the sequence (p_k) lives in the compact unit sphere. Compactness gives a convergent subsequence with limit p. The inequality $p_k \cdot x \leq p_k \cdot z_k$ for all $x \in C$, satisfied by each approximation, survives the limit because the inner product is continuous. The supporting hyperplane is the limit of approximating separating hyperplanes.

[1] Brouwer, of our fixed-point theorem had a very distinct opinion that proofs need to be "constructible," for—in his mind—they do not carry meaning otherwise. Hilbert, another famous mathematician, argued that this is like not allowing the boxer to use their fists. The fight between the two led to Hilbert (who was the editor-in-chief) removing Brouwer from the editorial board of the *Mathematische Annalen* a leading journal at the time.

The pattern is the same each time:

> Approximate → Apply the easy result → Extract a limit via compactness → Verify the limit inherits the property via closedness or continuity.

Recognizing this structure when you see it in a proof is half the battle.

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5. The economist's pattern book

Economic theorems cluster into a small number of recurring proof patterns. Knowing the pattern tells you, before reading a single line of a proof, what strategy to expect and which assumptions will be critical.

5.1 Existence via continuity and compactness

Pattern: The object exists because a continuous function on a compact set attains its value.

The archetype is the Weierstrass theorem (Topic 4). In economics, this pattern underlies: the existence of a utility-maximizing bundle (continuous utility, compact budget set), the existence of a profit-maximizing output (continuous profit function, bounded feasible set), and the existence of a best response in a finite game (continuous payoff in mixed strategies, compact simplex).

Why it works: Compactness supplies a convergent maximizing sequence; continuity passes the objective value through the limit. The proof is direct.

What breaks without the assumptions: Without compactness, the maximum may not be attained (a firm with unbounded output set and strictly increasing profit). Without continuity, the function may skip over its supremum (a payoff function with a jump discontinuity).

5.2 Existence via fixed point

Pattern: The equilibrium is a fixed point of the best-response map; existence follows from Brouwer or Kakutani.

This is the Nash theorem (Topic 10). The best-response correspondence maps strategy profiles to strategy profiles. Upper hemicontinuity comes from Berge's theorem (Topic 5): the argmax of a continuous function varies upper hemicontinuously with the parameters when the constraint correspondence is continuous. Convex values come from the linearity of expected payoffs in one's own mixing probabilities. Kakutani then delivers a fixed point.

Why the strategy is correct: The conclusion is existence of an equilibrium, which is an existence claim. Direct construction is not available (there is no formula for the equilibrium of a general game). Contradiction would require assuming no equilibrium and deriving a contradiction – possible but roundabout. Kakutani is the direct tool because it is exactly designed for this situation: self-maps of compact convex sets with well-behaved set-valued structure.

What each assumption buys: Compactness of the strategy simplex keeps the domain manageable. Convexity of the simplex is needed for Kakutani. Upper hemicontinuity (via Berge) ensures the best-response map is well-behaved. Convex values (via linearity of payoffs) is the assumption that fails if we restrict to pure strategies – which is why pure-strategy Nash equilibria need not exist in finite games.

5.3 Existence via order

Pattern: The fixed point (hence the equilibrium) exists because the best-response mapping is monotone on a lattice, and Tarski's theorem applies.

This appeared in Topic 10 and connects to the monotone comparative statics of Topic 8. In games with strategic complements – where higher play by others makes higher play by oneself more attractive – the best-response mapping is monotone increasing on the appropriate lattice. Tarski then gives both existence of a fixed point and the result that the set of equilibria is itself a lattice (with a largest and a smallest equilibrium).

Why this is useful beyond Kakutani: Tarski requires no compactness, no convexity in the geometric sense, and no continuity. Its currency is order and monotonicity. For models where the natural structure is an ordering (strategies that are "higher" or "lower" effort levels) rather than a convex set, Tarski is the right tool. It also gives comparative statics for free: if the environment shifts in a way that makes the best-response mapping shift up, the largest equilibrium shifts up too.

5.4 Uniqueness via contradiction

Pattern: Assume two solutions exist. Derive that they must be equal. Conclude there is only one.

The cleanest instance is the uniqueness part of the contraction mapping theorem (Topic 12): $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta\, d(x^*, y^*)$, which forces $d = 0$. The structure is: the two alleged solutions satisfy the same defining equation, and the equation's properties (here, the contraction property) are strong enough to force the distance between them to zero.

In consumer theory, the uniqueness of the optimal bundle under strict convexity of preferences follows the same structure: if two bundles were both optimal, the midpoint would be strictly preferred (by strict convexity), contradicting optimality of the original two.

Why uniqueness proofs are almost always by contradiction: Uniqueness is a *universal* statement ("there is no *other* solution"), and universal statements are most easily handled by assuming a counterexample (the second solution) and showing it cannot exist. A direct proof of uniqueness would need to show the solution is unique *without* referencing what a second solution would look like – which is awkward.

5.5 Characterization via necessary and sufficient conditions

Pattern: An object is characterized by a set of properties. The proof shows (a) any object with these properties satisfies the characterization and (b) any object satisfying the characterization has these properties.

This is the structure of the conditional expectation in Topic 15. The conditional expectation $\mathbb{E}[X|\mathcal{G}]$ is characterized as the unique $\mathcal{G}$-measurable random variable Z satisfying $\mathbb{E}[Z \cdot \mathbf{1}_A] = \mathbb{E}[X \cdot \mathbf{1}_A]$ for every $\mathcal{G}$-measurable event A. This is both a definition and a characterization: it says *exactly which property* pins down $\mathbb{E}[X|\mathcal{G}]$ uniquely, with no reference to a formula or an integral. The geometric interpretation – the conditional expectation is the orthogonal projection of X onto the subspace of $\mathcal{G}$-measurable functions – is a second characterization, and the proof that the two coincide is a proof that two sets of necessary and sufficient conditions describe the same object.

In mechanism design, the envelope theorem (Topic 7) characterizes how an agent's equilibrium utility changes with the agent's type and thereby links the allocation rule to the payment rule. In welfare economics, the first welfare theorem characterizes Pareto optimality: an allocation is Pareto optimal if and only if no feasible trade improves everyone. In each case, the value of the characterization is that it replaces a difficult search ("find all optimal allocations") with a testable condition ("check whether this condition holds at the candidate allocation").

First-order conditions are characterizations. The KKT conditions in constrained optimization (Topic 7) characterize the optimal solution under the conditions stated there: a constraint qualification gives necessity, while concavity of the maximization problem and convex feasibility give sufficiency. The proof therefore has two distinct halves. Both matter, and they use different tools – necessity uses the local structure of the constraint set, while sufficiency uses global shape restrictions.

5.6 Impossibility via contradiction

Pattern: Assume that the desired property holds. Derive a contradiction with an axiom or a structure theorem. Conclude the property is impossible.

Arrow's impossibility theorem is the canonical example in microeconomics (not developed in this pre-course but directly relevant). Assume a social welfare function satisfies unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship simultaneously. Then derive a contradiction. Each step of the proof involves showing that the axioms, jointly, force the social welfare function to resemble a dictatorship in some sub-domain, and then extending this until full dictatorship is established. The conclusion ("there is no such function") is existential negation, and proof by contradiction is the only available strategy – you cannot prove non-existence by construction.

The same structure appears in the proof that $\delta \geq 1$ breaks the contraction argument (Topic 12): with $\delta = 1$, the geometric series diverges and the Cauchy estimate fails. This is not an impossibility theorem in the Arrow sense, but it is the same logical structure: given an assumption (here, $\delta \geq 1$), the desired conclusion (convergence) cannot hold.

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6. Reading a proof in MWG

MWG's mathematical appendix and its main-text proofs are written in a compressed, symbol-dense style. Here is a protocol for reading one.

Step one: identify the claim. Before reading the proof, read the theorem statement carefully. Write down (or mark) what is *assumed* and what is *claimed*. These are the two endpoints of the chain you are about to read.

Step two: classify the strategy. Scan the first paragraph of the proof. Does it start "Suppose for contradiction…"? Then it is a contradiction proof, and you should look for where the contradiction appears. Does it start by constructing an object (a sequence, a set, a function)? Then it is probably a construction or approximation argument, and you should track what this object is and why it has the required properties. Does it start by writing out definitions and manipulating inequalities? Then it is a direct proof.

Step three: track the assumptions. Each hypothesis in the theorem statement should be used at least once in the proof. When you find a step that is not obvious, ask: which hypothesis justifies this? MWG proofs are tight enough that the answer is usually one specific assumption. If you cannot find it, that is a signal you have missed something – not that MWG made an error.

Step four: find the key step. Most proofs have one hard step and several easy ones. The hard step is usually the one where the most important assumption is used, or where an approximation is turned into an exact result, or where the contradiction is derived. Finding the key step is the same as finding the heart of the theorem. In the Kakutani proof, the key step is the closed graph argument that converts approximate fixed points into an exact one. In the IVT proof, it is the two-sided elimination of $f(\hat{x}) \neq c$. In the supporting hyperplane proof, it is the compactness of the unit sphere that gives the convergent subsequence of normals.

Step five: construct a counterexample for each hypothesis. For every hypothesis in the theorem, ask: if I drop this assumption and keep all others, does the conclusion still hold? If you can find a counterexample, you understand the theorem. If you cannot, you do not yet understand which direction of the argument needs the assumption. MWG occasionally provides counterexamples inline; the convergence theorems in Chapter 11 (and our Topic 15) explicitly include counterexamples for this reason.

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7. Writing a proof: how to start and how to get unstuck

Starting a proof from scratch is harder than reading one, because you do not know the structure in advance. Here is a small toolkit of starting moves.

Write down what you have and what you want. At the top of a scratch page: "Given: [hypotheses]. Want: [conclusion]." Spell out the definitions of each term in the statement. Often this step alone reveals the structure of the argument, because the definition of the conclusion is already partway to what you need.

Match the conclusion type to a technique.

- Conclusion is "there exists x with property P": try direct construction first. If you cannot construct x, try contradiction (assume no such x exists and derive something impossible) or approximation (build a sequence of approximate solutions and pass to a limit).
- Conclusion is "for all x, $P(x)$ holds": try direct proof (pick an arbitrary x and verify $P(x)$). If $P(x)$ is an inequality, try bounding one side by the other.
- Conclusion is "P implies Q": assume P and try to derive Q directly. If direct proof is stuck, try contrapositive (assume not-Q and derive not-P) or contradiction (assume P and not-Q and derive a contradiction).
- Conclusion is "P and Q are equivalent": prove both $P \Rightarrow Q$ and $Q \Rightarrow P$ separately, by any technique.
- Conclusion is "there is exactly one x with property P": prove existence (see above) and uniqueness (assume two, derive they are equal, usually by contradiction).

If stuck on a direct proof, introduce the right intermediate object. Most proofs require something that is not in the hypotheses: a cleverly defined set (as in the IVT proof), a norm (as in the contraction proof), a sequence of approximations (as in Kakutani), or an auxiliary function (as in the Brouwer 1D proof, where $g(x) = f(x) - x$ converted a fixed-point question into a zero-finding question). There is no algorithm for finding this intermediate object. But the object is almost always motivated by what the conclusion requires: ask "what would I need in order to apply [the relevant tool]?" and define the intermediate object to supply it.

Use the hypotheses. If a proof is going nowhere, check whether you have used every hypothesis. Theorems are stated with exactly the assumptions needed. If you have an unused assumption, it almost certainly points to the missing step.

Work backwards. Start from the conclusion and ask "what would I need in order to conclude this?" Then ask "what would I need in order to have *that*?" Keep working backwards until you reach the hypotheses. If the forward and backward chains overlap, you have the proof.

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8. A worked example

We close with an example that is small enough to be followed completely but rich enough to illustrate several techniques at once. It also has an immediate economic interpretation.

Claim. Let $f : [0,1] \to [0,1]$ be continuous and strictly decreasing. Then f has exactly one fixed point.

Economic interpretation. In a Cournot duopoly, suppose each firm's best response to the other's quantity is strictly decreasing (strategic substitutes: when my rival produces more, my optimal output falls). If the best-response functions map $[0, \bar{q}]$ to $[0, \bar{q}]$ and are continuous, this claim guarantees a unique Nash equilibrium.

Step 1: Existence (direct proof using the IVT)

Define $g : [0,1] \to \mathbb{R}$ by $g(x) = f(x) - x$.

Then $g(0) = f(0) - 0 = f(0) \geq 0$ (since $f(0) \in [0,1]$) and $g(1) = f(1) - 1 \leq 0$ (since $f(1) \in [0,1]$).

Since g is continuous (as the difference of two continuous functions – recall Topic 4) and $g(0) \geq 0 \geq g(1)$, the Intermediate Value Theorem (Topic 4, §7) gives a point $x^* \in [0,1]$ with $g(x^*) = 0$, i.e., $f(x^*) = x^*$.

Why this strategy? A fixed point is a zero of $g(x) = f(x) - x$. The IVT is the canonical tool for finding zeros of continuous functions on intervals. The reduction $f(x) = x \Leftrightarrow g(x) = 0$ is the key intermediate step, and the boundary conditions $g(0) \geq 0 \geq g(1)$ follow immediately from the assumption that f maps $[0,1]$ into itself.

Step 2: Uniqueness (proof by contradiction)

Suppose x^* and y^* are both fixed points with $x^* < y^*$. Then $f(x^*) = x^*$ and $f(y^*) = y^*$.

Since f is strictly decreasing and $x^* < y^*$, we have $f(x^*) > f(y^*)$. But $f(x^*) = x^*$ and $f(y^*) = y^*$, so $x^* > y^*$. This contradicts $x^* < y^*$.

Therefore no two distinct fixed points can exist.

Why contradiction? Uniqueness claims are universal statements ("there is no second fixed point"), and the natural move is to assume a second fixed point and show it leads to something impossible. Here the contradiction is immediate: the strict decrease of f forces $f(x^*) > f(y^*)$, but the fixed-point property forces the reverse inequality.

Which assumption did the work? Existence used continuity (to apply the IVT) and the self-mapping property (to get the right signs at the boundary). Uniqueness used strict decrease. If f were only weakly decreasing, there could be an interval of fixed points. If f were constant, every point in the interval would be a fixed point. Strict decrease is exactly what prevents this.

Summary of techniques used:

Step	Technique	Key assumption used
Reduction to g's zero	direct manipulation	definition of fixed point
Boundary signs	direct verification	$f : [0,1] \to [0,1]$
Existence	IVT (itself a direct proof)	continuity of f
Uniqueness	contradiction	strict decrease of f

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9. Where this appears in MWG and beyond

MWG does not have a chapter on proof techniques as such. The relevant background is spread across:

- **MWG, Mathematical Appendix A** (sets, functions, relations – the language in which proofs are written),
- **MWG, Mathematical Appendix C** (fixed-point theorems stated, without proof),
- **MWG, Chapter 16 and 17** (welfare theorems and general equilibrium – where the separating hyperplane and Kakutani appear as working tools, not as pure mathematics).

For a longer treatment of proof technique oriented toward economists, Efe Ok's *Real Analysis with Economic Applications* (2007) is the standard reference. It covers all the theorems in this pre-course with full proofs and extensive economic motivation. Chapter A (logic and sets) is a good place to go if you want to sharpen the formal side of what this topic covers informally.

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10. What you should take away

After this topic, you should be able to:

- identify the proof technique (direct, contradiction, contrapositive, induction, construction, approximation-then-limit) being used in a proof you are reading,
- explain, for each technique, which features of the conclusion or the hypotheses make it the appropriate choice,
- locate the key step in a proof and identify which hypothesis it uses,
- recognize the five recurring patterns in economic proofs (existence via fixed point, existence via order, uniqueness via contradiction or contraction, characterization via necessary and sufficient conditions, impossibility via contradiction),
- and start a proof of your own by matching the conclusion type to a technique and working forward from the hypotheses or backward from the conclusion.

You will not write novel existence proofs on a problem set. But you will write uniqueness arguments, verify first-order characterizations, check that a given mapping satisfies a fixed-point theorem's hypotheses, and follow multi-step proofs in MWG without getting lost. The techniques in this topic are the tools for all of that.